Name: _____

Radioactivity 4 H

Date:

Time:

Total marks available: 38

Total marks achieved: _____

Questions

Q1.

Figure 5 shows a PET scanner used to detect cancerous tumours.

A radioactive isotope is injected into a patient.

The isotope is absorbed by the tumour.

The isotope emits positrons from the location of the tumour.

The ring of radiation detectors rotates around the person.

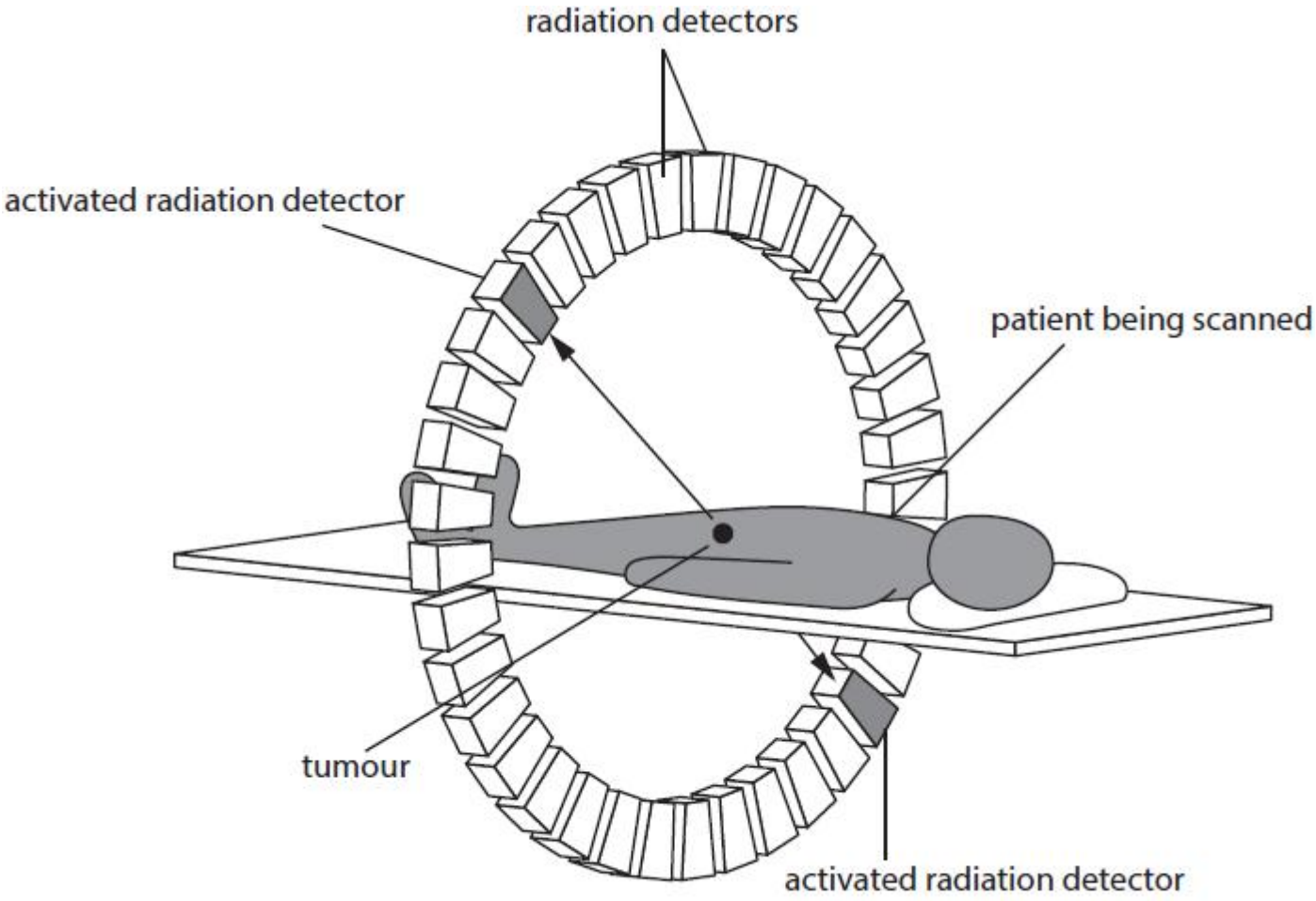

Figure 5

(i) Explain how the scan can give the location of the tumour.

(3)

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(ii) Explain why the radioactive isotope injected into the patient must be produced near to the place where it is to be used.

(2)

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(Total for question = 5 marks)

Q2.

A teacher determines the background radiation count rate in a laboratory.

Explain how to determine a value for the background radiation count rate.

(3)

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(Total for question = 3 marks)

Q3.

Early in the twentieth century, scientists fired a beam of alpha particles at thin gold foil.

Figure 2 shows the main parts of their experiment with some results.

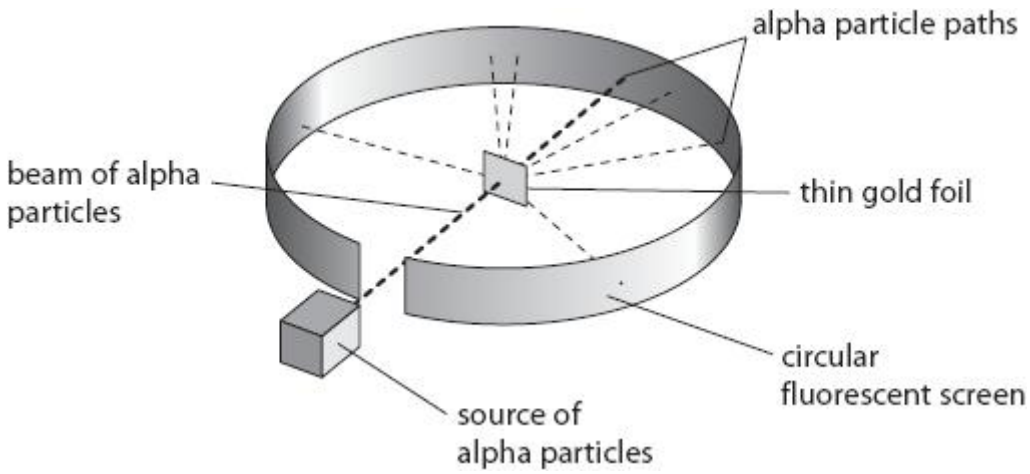

Figure 2

Explain how the results of the experiment shown in Figure 2 support the nuclear model of an atom.

(3)

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(Total for question = 3 marks)

Q4.

Explain how neutrons enable a nuclear chain reaction to take place.

(2)

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(Total for question = 2 marks)

Q5.

Some isotopes are unstable.
They emit β^- particles when they decay.

Explain how a nucleus changes when a β^- particle is emitted.

(2)

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(Total for question = 2 marks)

Q6.

The apparatus that was used in the experiment is shown in Figure 10.

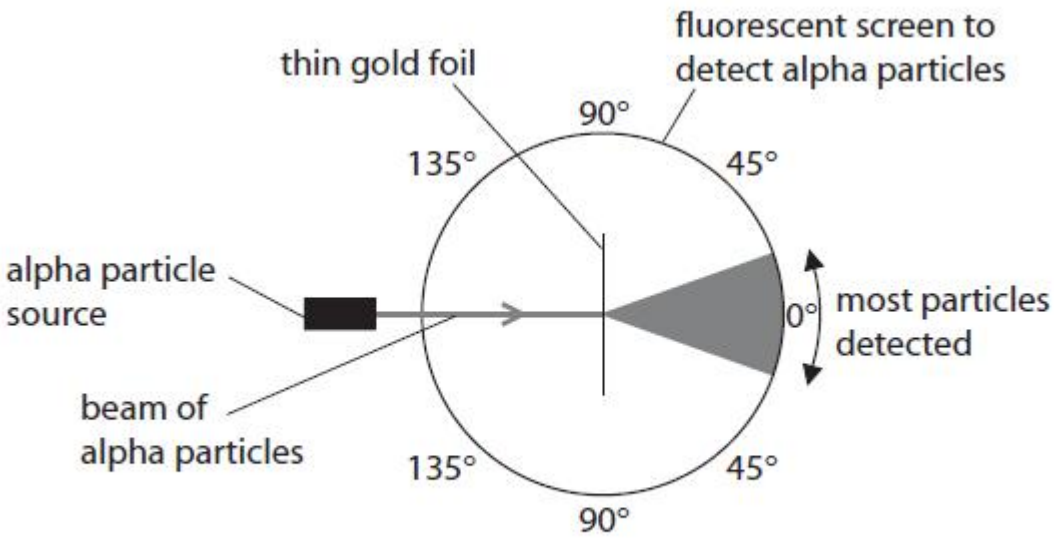

Figure 10

(i) The number of particles detected at each angle in a given time is shown on the graph in Figure 11.

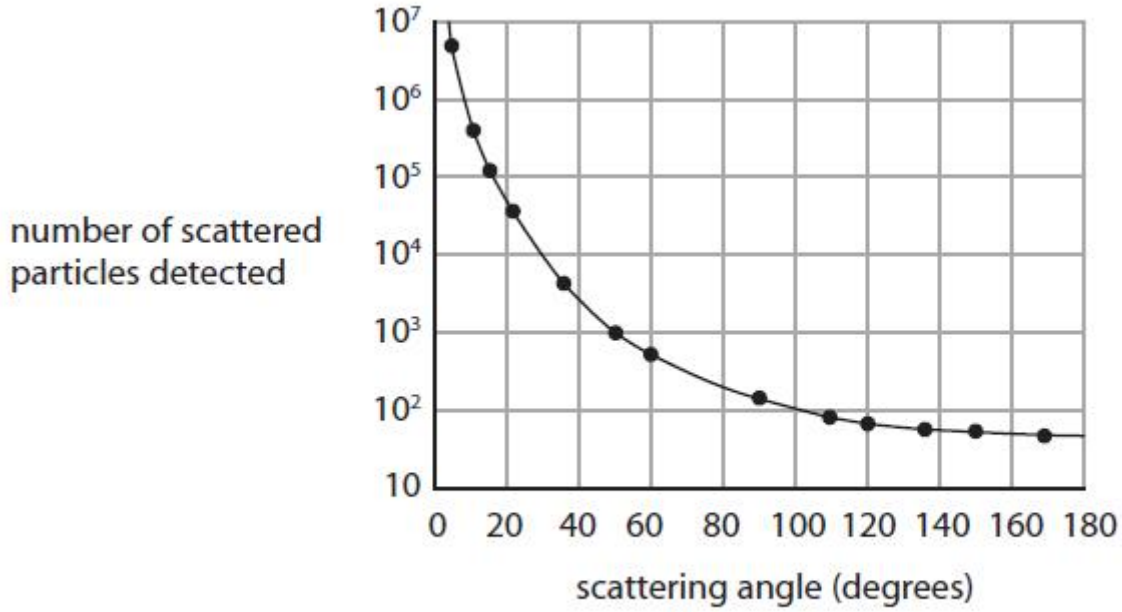

Figure 11

Use information from the graph.
Estimate the ratio of the number of particles scattered through 5° to the number of particles scattered through 100°.

(2)

ratio = ...

(ii) Explain how the difference in the number of particles scattered at different angles gives evidence for the current model of the structure of the atom.

(4)

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(Total for question = 6 marks)

Q7.

Figure 6 is a diagram of a nuclear reactor.

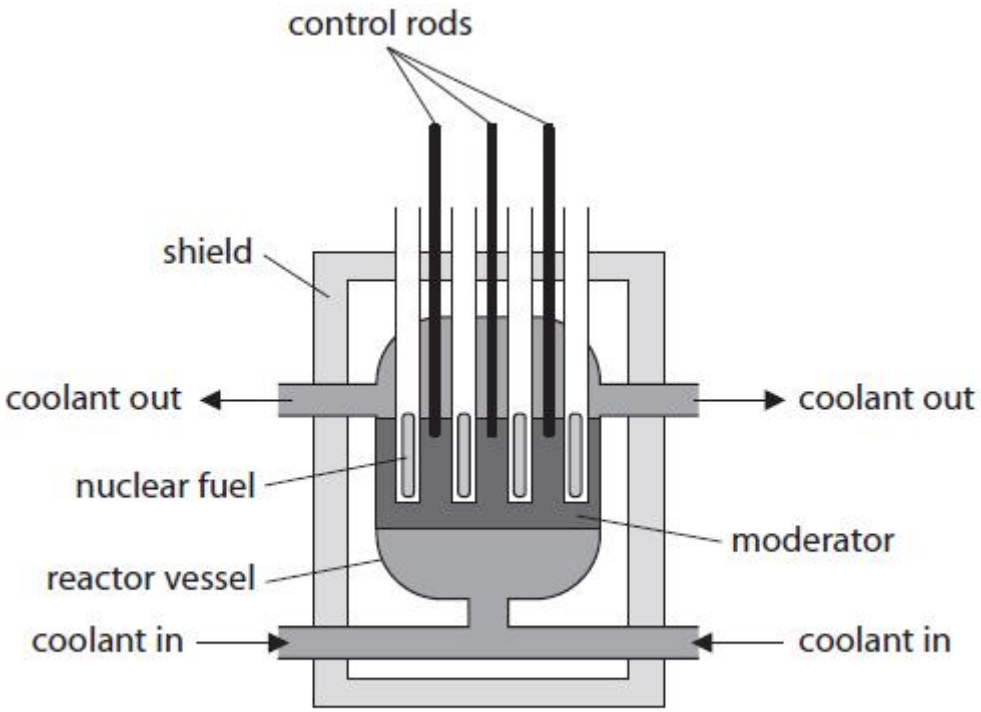

Figure 6

(i) Explain how pushing the control rods further into the reactor slows down the nuclear chain reaction.

(2)

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(ii) The moderator in a nuclear reactor slows down the neutrons so that the neutrons are more likely to start other fission reactions.

In a nuclear reactor,

- the average speed of the fast neutrons is 3.0 × 107 m/s
- the average speed of the slow neutrons is 4.0 × 103 m/sCalculate the average speed of the slow neutrons as a percentage of the average speed of the fast neutrons.

(2)

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(iii) The nuclear reaction is the first stage in the process of generating electricity.

Describe how energy is transferred from the nuclear reaction to the next stage in the process.

(2)

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(Total for question = 6 marks)

Q8.

In a nuclear reactor, a chain reaction is produced and controlled.

Explain what happen inside a nuclear reactor if neutron speeds are not controlled.

(3)

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(Total for question = 3 marks)

Q9.

Explain the use of a moderator in a nuclear reactor.

(3)

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(Total for question = 3 marks)

Q10.

Carbon–14 decays into nitrogen–14.

The symbol for nitrogen–14 is $^{14}_{7}N$

Explain what happens in a carbon-14 nucleus when it decays to a nitrogen-14 nucleus.

(2)

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(Total for question = 2 marks)

Q11.

Figure 8 shows what happens during a fission chain reaction.

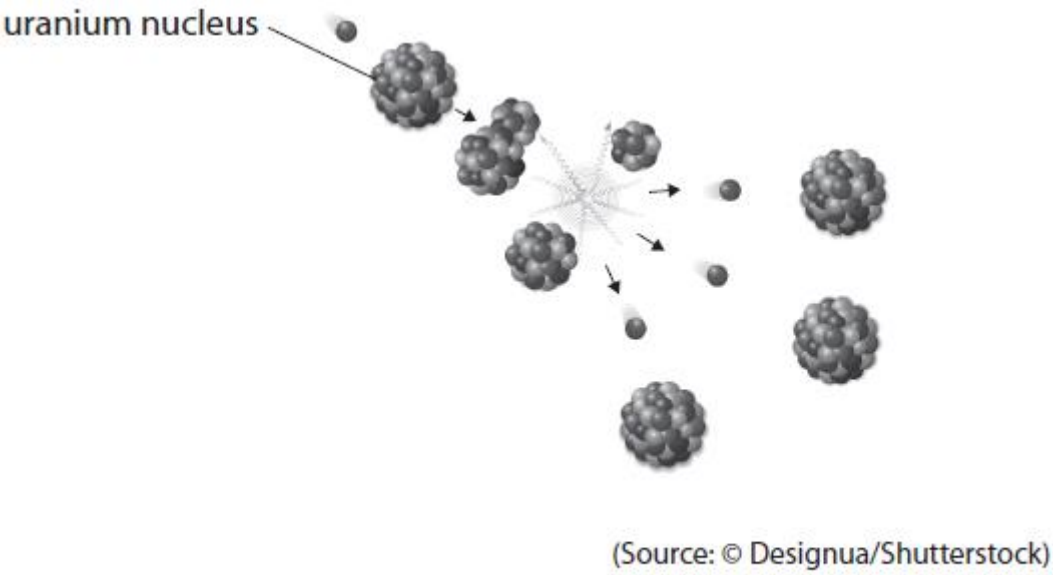

Figure 8

Explain what happens when **one** neutron interacts with **one** uranium nucleus.

(3)

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(Total for question = 3 marks)